

Exercise 1 – Language identification: from linear to deep

Deadlines

Deadline for Exercise 1: **Sunday, 11.10.2026, 23:59 (Zurich Time)**.

Peer review starts: Monday, 12.10.2026, 20:00 (Zurich Time).

Deadline for the peer review: **Monday, 19.10.2026, 23:59 (Zurich Time)**.

- Note, the peer reviews will be assigned one day after the exercise deadline. You will find instructions for the peer review process at the end of this document.

Learning goals

This exercise consists of two parts: the first part aims at deepening your understanding of linear models. The second part will already target a simple kind of multi-layered network; the [multilayer perceptron](#) (MLP). Don't worry if you don't know anything about MLPs when reading. We will cover all you need to know to solve the second part of this exercise next week in class and in the tutorial session. The learning goals of this exercise are...

- to understand linear models and use them for [multiclass classification](#) tasks.
- to be able to implement different machine learning models, from linear models in [scikit-learn](#) to a multilayer perceptron (MLP).
- to understand the role of hyper-parameters and regularization.
- to perform an error analysis of machine learning models.
- to observe the most important features that lead to a prediction of a specific class and gain some insights into how the model makes its predictions
- to conduct an ablation study.

Please keep in mind that you can always consult us and use the [exercise forum](#) if you get stuck (note that we have a separate forum for the exercises).

Deliverables

Your solutions for each part should be submitted as self-contained iPython notebooks (ipynb files). The notebooks should contain your well-documented, EXECUTED, and EXECUTABLE code. That way, your reviewers can view and execute your code without potential dependency issues and/or installing new packages or versions of packages. We encourage you to solve the exercise on Google Colab and download the .ipynb file when everything is completed.

You will also have to write a short lab report at the end of each script.

The lab report should be written in a markdown cell and contain a detailed description of the approaches you used to solve this exercise. You can discuss what worked and what didn't work, but in both cases, you please provide reasoning as to why something did or didn't work. Please also

include the results. Inside each notebook, there are specific questions denoted with “ ?” that should be addressed in the report.

Please submit a zip folder named **ex1.zip** containing the following working ipynb notebook scripts, named as follows:

- **ex1_lr.ipynb**
- **ex1_nn.ipynb**

Please note:

- Organise your code such that it is executable.
- The cells must have already been run and the output must be visible to anyone checking your notebook without having to run the code again.
- Your assessors must be able to run your code. If it doesn't work, you can't get the maximum score.
- Please make sure that your answers to the specific questions outlined in the notebook and denoted with “ ?” are clearly marked. If the assessors cannot easily find the answers they need, you can't get the maximum score. We recommend including these clearly in your lab report.
- Complete the two declarations at the end of each notebook (contribution declaration and declaration on the use of AI). The rules on the use of generative AI for this exercise are stated at the top of the notebooks.
- Also, we highly recommend testing that your notebook runs as expected before submitting it. To do this, hit “Runtime” > “Restart runtime and run all”, and make sure that there are no errors!

Data

For both parts of this exercise, you will work with the same data. The goal is to classify languages based on text snippets. This is a challenging extension of the problem described in Goldberg: chapter 2. However, we will work with more languages than just six and the text segments we need to classify are a bit shorter. In the notebooks provided as starting points, there are download links directly to the data, which will allow you to fetch the raw datasets.

Part 1 - Language identification with linear classification

To facilitate the start, you can use [this iPython notebook](#) in Google Colab which loads the files using the public links. We recommend making a copy of the notebook to build on. Simply hit the “Copy to Drive” button to do this.

The aim of part 1 is to become familiar with logistic regression, a linear classification model, as well as with the basic functionality of sklearn, a useful Python library for machine learning tasks.

Your task is to work through the notebook provided and implement the necessary code to:

1. Explore the data and create training/test splits for your experiments
2. Build a [LogisticRegression](#) classifier and design some relevant features to apply it to your data
3. Conduct hyperparameter tuning to find the optimal hyperparameters for your model

4. Explore your model's predictions and conduct an error analysis to see where the model fails
5. Conduct an analysis of most important features used by the model to make its predictions
6. Conduct an ablation study using a subset of languages

Part 2 - Your first Neural Network

Now that you've got some experience under your belt, let's see if you can beat the best linear model you've trained with sklearn with a simple **neural network**, a multilayer perceptron (MLP). Again, we provide you with [this iPython notebook](#) to get you started.

Your task is to work through the notebook provided and implement the necessary code to:

1. Build a simple multilayer perceptron classifier.
2. Train your neural network classifier on a GPU.
3. Try to improve the model's accuracy above 87%.

Submission & Peer Review Guidelines:

For all of the exercises, peer reviews will be carried out on OLAT.

The peer review phase starts on the day after the submission deadline (see Deadlines) and ends at the peer review deadline. During this time you review the submissions of your peers. You need to do **1 review per person** to get the maximum points for this exercise.

Here are some additional rules:

- **All file submissions are anonymous (for peer review purposes): Do not write your name into your Jupyter Notebooks.**
- **EVERY** member of each team submits on OLAT.
- Please submit a zip folder containing all the deliverables.

Groups & Peer Reviews:

- You should work in the same group for all exercises. Each member should contribute equally!
- Before the substantive review, check that both declarations are present and completed. You are not asked to verify whether they are true.
- **If you do not submit your peer review, you will receive a -0.25 point penalty on the final grade of the exercise.**
- Please use full sentences when giving feedback. Please write the peer review in English.
- Provide **constructive** and helpful comments to your peers! If you criticise something, you should also be able to provide actionable and realistic suggestions for how something can be improved.
- Please answer all the pre-designed review questions of the peer review to the best of your ability.